

Geocomputation & Machine Learning course

Quick recap

The meeting was a lecture led by Giuseppe on using the Linux operating system and Bash commands for data management. Giuseppe checked that all participants had their virtual machines running and shared folders set up, and then explained the basics of Linux distributions, package installation with APT, and the structure of the file system. He demonstrated how to navigate directories with commands like LS, CD, and PWD, and emphasized the use of tab completion and keyboard shortcuts to work efficiently. The session covered the difference between text and binary files, file redirection with `>` and `>>`, and the use of commands like WC, GREP, FIND, SORT, MORE, LESS, and RM. Giuseppe explained special characters and regular expressions for pattern matching, and showed how to use pipes to chain commands. He also introduced Bash variables, for loops, and how to use them for batch processing of text files, including examples with climate data. Participants like Abdul, Sebastián, and others asked questions about specific commands, shortcuts, and file formats, and Giuseppe provided clarifications and practical tips throughout the session.

Next steps

Abdul

- Practice Bash commands, especially loops and conditions, over the weekend to gain expertise.

Ali

- Install the virtual machine and resolve the system problem, aiming to fix it today.

Andrea

- Upgrade Ubuntu to 26.04 LTS as soon as possible, preferably during the weekend.

- Clone the course repository using git clone and work in a separate folder (e.g., SEData) rather than the cloned folder.

Sebastián

- Operate on CSV files by converting commas to spaces using the ``tr`` command, then manipulate them with Bash.

Collaboration

- All participants: Practice navigating the Linux terminal using commands (ls, cd, pwd, etc.) and utilize tab completion and keyboard shortcuts (Ctrl+A, Ctrl+E, Ctrl+R, Ctrl+L) to become faster.
- All participants: Practice manipulating text files using Bash commands (grep, wc, sort, find, redirection, piping) and experiment with for loops and variables.
- All participants: Prepare to continue working in the Bash environment and get familiar with it before the next lecture on AWK.

Summary

Linux Terminal Fundamentals Training

Giuseppe led a technical session focused on Linux operating system fundamentals and terminal navigation. He explained Linux distributions, the Bash shell syntax, and important commands for directory navigation including LS, CD, and LL. The session covered practical terminal skills such as file completion using the tab key and proper directory structure navigation without spaces in file names. Giuseppe emphasized the importance of using the terminal for managing geospatial data and preparing for future HPC work, encouraging participants to practice these skills extensively.

Terminal Commands and File Navigation

Giuseppe explained the use of tab completion in the terminal and demonstrated how to redirect terminal output to text files. He distinguished between text files (which are human-readable) and binary files (which require special software to interpret), using examples like TIFF

and CSV files. The class also covered commands for viewing text files including "more" and "less", as well as the "find" command for searching directories, and discussed keyboard shortcuts for command history navigation using arrow keys.

Terminal Keyboard Shortcuts Demo

Giuseppe demonstrated keyboard shortcuts for the terminal, including Ctrl-A for selecting text and Ctrl-E for moving to the end of a line. He explained the important Ctrl-R shortcut for reverse search through previously executed commands, allowing developers to go back in time and edit previous commands. Giuseppe also covered file redirection techniques, noting that using ">>" instead of ">" allows commands to append output to files rather than overwrite them.

Terminal Commands Training Session

Giuseppe provided guidance on using terminal commands in both Mac and Linux environments, explaining that Mac terminals use bash syntax similar to Linux. He demonstrated file management commands including LS for listing files, RM for deleting files, and special characters like asterisks for wildcard searches. The discussion concluded with an explanation of piping, showing how to use the pipe symbol to direct command output as input for subsequent commands, with Giuseppe noting that piped commands are faster than writing intermediate files to disk.

RAM vs Disk Storage Overview

Giuseppe explained the differences between working with RAM and disk storage, noting that RAM operations are faster but have limitations on file size compared to disk storage. He demonstrated how to work with text files using command-line tools like head, grep, and word count, and showed how to manipulate and explore data within specific directories. Giuseppe emphasized the importance of understanding directory structures and correctly specifying file paths to avoid common mistakes. The session concluded with a question about how to extract specific data columns from the text files, which remained unanswered.

Bash Text Processing Commands

Giuseppe demonstrated various text processing commands in Bash, including using grep with spaces and quotation marks for precise searching, and explained sorting operations with examples of sorting by columns and handling binary versus decimal formats. He covered the use of variables in Bash scripts, showing how to assign values and print them, and explained the syntax for for loops with practical examples of iterating through years and counting observations. The discussion concluded with a comparison of two approaches to file processing within for loops, highlighting differences in when data is saved during versus after the loop execution.

Bash File Handling Techniques

Giuseppe explained the differences between reading and writing to files in Bash, emphasizing the importance of file handling efficiency and memory management. He demonstrated how to use grep for single-pass file operations and highlighted debugging techniques for for loops. Abdul expressed interest in practicing these commands over the weekend, particularly focusing on loops and conditions, while Sebastián inquired about working with CSV files. Giuseppe advised on handling CSV files and suggested using the tr command to convert comma separators to spaces for easier Bash operations. The group agreed to continue learning Bash and AWK in next week's sessions on Tuesday and Thursday.